

# WENTWORTH LIU

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Data Scientist · Data Automation & Model Workflows · Scientific Computing · Operational Analytics

## SUMMARY

Data Scientist with 2+ years of experience building automated, data-driven workflows for complex engineering and operational systems. Strong in Python and SQL, with experience in multi-source data integration, validation, transformation, configurable feature systems, and reusable analytical workflows for million-scale EV fleets at Li Auto. Brings a Computational Materials Science background and a careful, automate-first approach to improving data quality, consistency, and decision-support outputs.

## EXPERIENCE

**Li Auto** Beijing, China  
Data Scientist, Battery Intelligence Department Jul 2024 – Present

- **Automated Data Transformation & Model Workflows.** Built a configurable feature system for multi-dimensional fleet telemetry using custom event definitions, multi-operator aggregations, and cross-platform normalisation; deployed reusable workflows to an internal ML platform and improved feature-iteration efficiency by ~85%.
- **Data Validation & Analytical Outputs.** Built an explainable risk-ranking pipeline across multiple battery platforms, producing vehicle-level risk scores and feature-level drivers; achieved 93% Recall@2% on retrospectively validated high-risk cases while limiting review to the highest-risk 2% of vehicles.
- **Scenario Analysis & Decision Support.** Used quasi-causal analysis on million-scale telemetry to identify high-risk SOH-degradation cohorts and actionable drivers; productised intervention analysis across multiple vehicle models to support consistent scenario assessment and targeted decisions.
- **Workflow Automation.** Built a hybrid rule-based and statistical classifier for battery-quality work orders across 30+ fault categories, combining text matching, TF-IDF, and Random Forest; achieved Weighted F1 > 0.90 and used confidence-based routing to focus review on ambiguous cases.

**ByteDance** Beijing, China  
Data Analytics Intern, Douyin Market Strategy May – Sep 2023

- **Multi-source Integration & Data Monitoring.** Integrated consumption, dissemination, and search signals across multiple data platforms into an IP popularity measurement system; added automated pipeline monitoring to identify upstream data instability and support reliable recurring analysis.
- **Metric Design & Analytical Reporting.** Redesigned performance metrics and built a multi-dimensional content taxonomy for Qishui Music; used funnel segmentation and audience-saturation modelling to diagnose conversion decline and inform content and resource-allocation decisions.

**Dongwu Securities** China  
Equity Research Intern, Mechanical Sector Dec 2021 – Mar 2022

- **Quantitative Analysis & Reporting.** Combined company fundamentals, competitive analysis, and market dynamics with stock-price modelling to deliver structured investment recommendations for Harmonic Drive Systems.

## EDUCATION

**Nanjing University** Sep 2021 – Jun 2024  
MSc Computational Materials Science

- Developed Skeaf, a 3D Fermi-surface visualisation tool combining HDBSCAN clustering with quantum-oscillation data; first author, Physical Review B (2024); co-author, Physical Review B (2024) and Physical Review Materials (2023); Patent: CN115795266A.

**Nanjing University** Sep 2017 – Jul 2021  
BSc Materials Science & Engineering (Outstanding Graduate) · GPA: 4.6/5.0 · Top 1% of cohort

- National Scholarship (top 0.15%) · MCM/ICM 2020 Honorable Mention (Team Captain) · Sakura Science Exchange Programme, Tohoku University, Japan (5 selected university-wide).

## SKILLS

|                                 |                                                                                                                                                                                              |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Programming</b>              | Python · SQL/HiveSQL · Pandas · NumPy · C++ · MATLAB · Git                                                                                                                                   |
| <b>Data Automation</b>          | ETL Pipelines · Data Validation · Data Transformation · Multi-source Data Integration · Data Standardisation · Automated Monitoring · Workflow Automation · Reporting · Visualisation        |
| <b>Analytics &amp; Delivery</b> | Statistical Modelling · Predictive Modelling · Machine Learning · Feature Engineering · Time Series · Causal Inference · Data Productisation · GCP · Docker · CI/CD · Spark · Hadoop · Linux |
| <b>Domain &amp; Lang.</b>       | Scientific Computing · Engineering Data · EV Analytics · English (TOEFL 104) · Mandarin/Cantonese (Native) · Japanese (JLPT N1)                                                              |